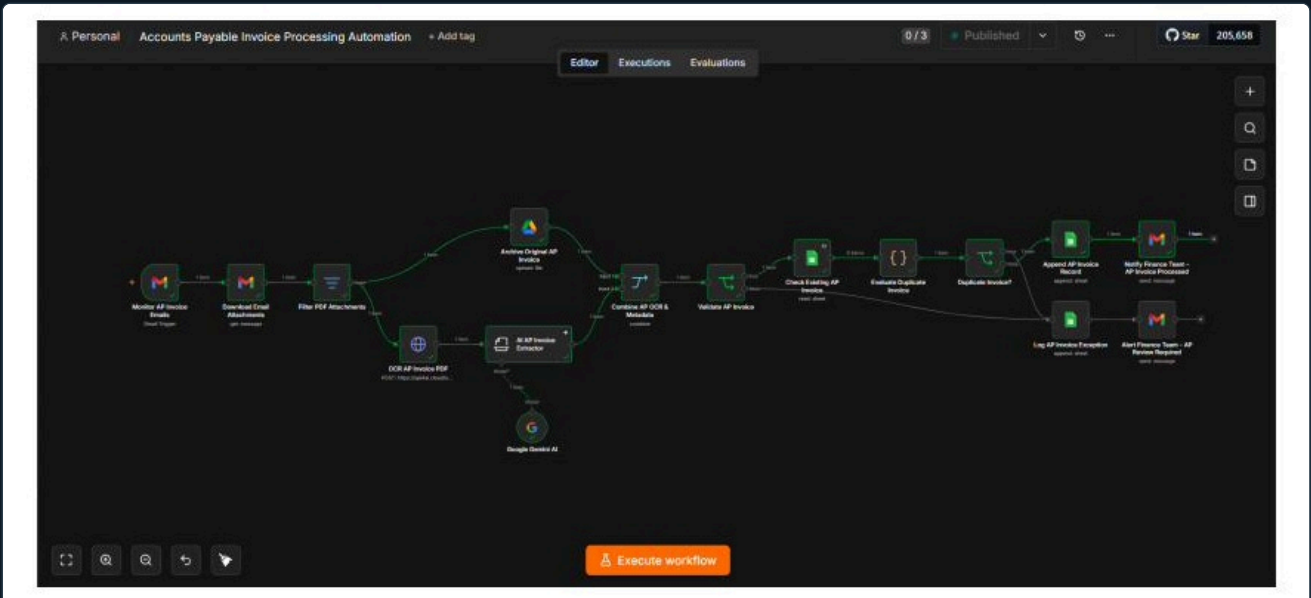


Accounts Payable Invoice Processing Automation

From Inbox to Verified Record — Automatically

Every invoice that reaches this inbox is read, checked against existing records, and resolved to one of two outcomes — a Gmail-triggered automated pipeline for accounts payable teams that replaces manual transcription and inconsistent review with a system where every processed invoice lands in either a verified register entry or a flagged, reviewable exception.

Industry	SMB Finance & Accounts Payable Teams · Accounting & Bookkeeping Firms
Automation Type	Invoice Intake, AI Extraction, Validation & Duplicate Detection
Built With	n8n · Gmail · Google Drive · Google Sheets · Google Gemini AI · api4ai OCR
Primary Objective	Replace manual invoice transcription with a pipeline where every processed document reaches a recorded outcome



The live, published n8n build — current workflow canvas.

The Problem

Manual invoice intake follows the same pattern in nearly every AP team: an email arrives, someone opens the PDF, retypes the numbers into a spreadsheet, files the original somewhere, and moves to the next one. At low volume it's tedious but survivable. At scale, it's a liability.

Every transcription step is a place for a digit to be misread or a due date mistyped — errors that surface later as payment problems, not at the point they're created. Originals sit in inboxes with no consistent archive, and review standards shift from person to person. When a document is incomplete, unreadable, or already paid once, the usual outcome isn't a flag — it's silence.

What AP teams need at intake isn't more manual review. It's a consistent front door: one that reads every document the same way, checks it against what's already on file, and never lets it disappear without a trace.

The Solution

This pipeline is that front door — a Gmail-triggered automated pipeline that begins working the moment an invoice email arrives.

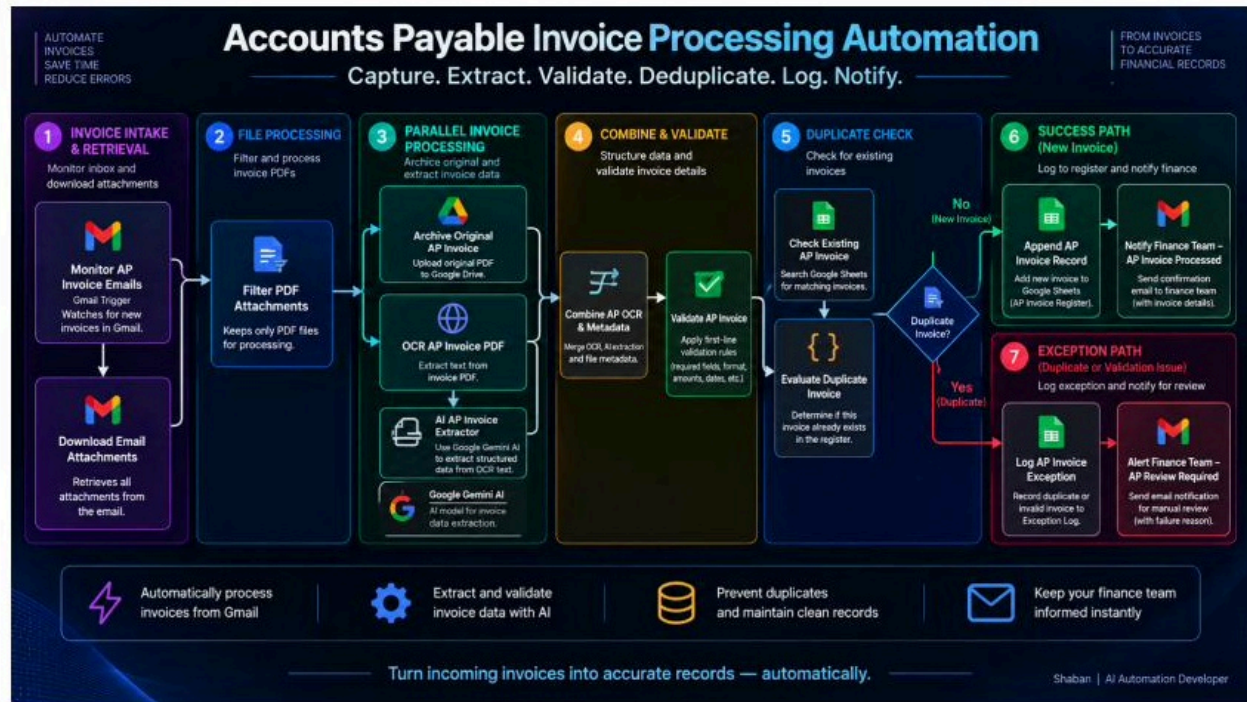
It's built on a firm separation of responsibility. Deterministic logic makes every decision that can be defined by a rule: whether a required field is present, whether an amount is valid, whether this invoice already exists in the register. AI is scoped to a single, narrow task — reading unstructured OCR text into structured data. It doesn't decide outcomes; it prepares the record the rules act on.

The overall processing architecture follows the same eight stages for every invoice: Receive → Archive → OCR → Extract → Validate → Deduplicate → Record → Notify or Escalate — with validation failures routed directly to the exception path ahead of the duplicate check.



How It Works

The diagram shows the full path from inbox to record. Three engineering decisions define how it actually behaves:



Current workflow architecture — seven stages, intake through exception and success paths.

01

Parallel archival and OCR

Archival and OCR run in parallel, not in sequence. The moment a PDF is downloaded, it's uploaded to Google Drive and sent for OCR at the same time — the original preserved independently of whatever happens downstream.

02

Validation before duplicate detection

Validation happens before duplicate detection, not instead of it. A document must clear field-level checks — invoice number, vendor name, amount, due date, OCR quality — before the workflow compares it against existing records. Each failure is named specifically, not returned as a generic error.

03

Duplicate check before record creation

The duplicate check runs immediately before a record is written, not after. Only an invoice that has already passed validation is checked against the AP Invoice Register used by the workflow — and only if it clears that check does a new record get created.

Everything else in the diagram — trigger, attachment filter, AI extraction, metadata merge — exists to feed these three decisions cleanly.

Demonstrated Evidence

The clearest evidence of how this pipeline behaves is what happened when the same invoice arrived twice.

Shared identifiers across both submissions — **INV-TEST-001** · **Acme Office Supplies Ltd** · **\$1,180.00**

First submission

Invoice → Validation → Duplicate Check → AP Invoice Register → Success Notification

Invoice INV-TEST-001 — Acme Office Supplies Ltd, \$1,180.00 — passed field validation, cleared the duplicate check against an empty register, and was appended as a new record. A confirmation email reached the finance team, citing the same number, vendor, and total.

Transaction	Invoice Number	Vendor Name	Vendor Email	Vendor Address	Vendor Phone	PO Number	Invoice Date	Due Date	Payment Terms	Currency	Subtotal	Tax Amount
1	INV-TEST-001	Acme Office Supplies Ltd	finance@acme.com	1100 Commerce Bldg	555-123-4567	PO-1234	2026-09-20	2026-10-20	Net 30	USD	1180.00	0.00

AP Invoice Register — row 2 is the new record created on first submission.

Invoice Processed Successfully

Hi Finance Team,

This AP invoice passed the configured deterministic intake validation rules and was written to the AP Invoice Register.

Invoice Number	INV-TEST-001
Vendor	Acme Office Supplies Ltd
Invoice Date	2026-09-20
Due Date	2026-10-20
Total Amount	1180.00
Processing Status	Valid - Processed

[Open Archived Invoice](#) [Open AP Invoice Register](#)

Confirmation email sent to the finance team.

Second submission

Same Invoice → Duplicate Check → AP Exception Log → Review Notification

The same invoice — same number, same \$1,180.00 total — arrived a second time. It cleared the same first-line validation, but this time the duplicate check matched it against the existing register entry and stopped it: no second row was created. It was logged to the AP Exception Log instead, Failure Reason "Duplicate Invoice," and a review notification replaced the confirmation.

Transaction	File Name	File Link	Failure Reason	Vendor Name	Invoice Number	Extracted Amount	Review Status	Notes / Actions	How OCR Test
1	INV-TEST-001-001	INV-TEST-001-001	Missing Invoice Number	Acme Office Supplies Ltd	INV-TEST-001	1180.00	Pending Review	Review this entry	Failed
2	INV-TEST-001-002	INV-TEST-001-002	Duplicate Invoice	Acme Office Supplies Ltd	INV-TEST-001	1180.00	Pending Review	Duplicate invoice	Failed

AP Exception Log — row 3, TEST-003-Duplicate, matches INV-TEST-001 at \$1,180.

AP REVIEW REQUIRED

Invoice Processing Exception

The invoice was archived, but it did not pass the configured first-line AP validation or duplicate check. It has been routed to the exception log for manual review.

Vendor	Not Found
Invoice Number	Not Found
Extracted Amount	Not Found
Failure Reason	Missing Invoice Number: Missing Vendor Name: Invalid Total Amount: Invalid Due Date: OCR Quality Failure
Review Status	Pending Review

[Open Archived Invoice for Review](#)

Exception alert template — the same notification fires for duplicates and validation failures alike. *Shown here is a separate validation-failure example, not the literal alert for this duplicate.*

No second AP Invoice Register entry exists for INV-TEST-001. The register, the exception log, and both emails independently agree on the same invoice number and amount — what makes this a verifiable outcome, not a described feature.

Secondary evidence — same Exception Log

The exception log also shows validation catching incomplete documents on its own terms — one invoice logged as "Missing Invoice Number," another as "Missing Vendor Name" — each naming the specific gap rather than a generic rejection.

What this evidence covers. The workflow currently implements Gmail-triggered intake, parallel archival and OCR, AI-assisted extraction, multi-field validation, and duplicate detection against the AP Invoice Register — shown here with synthetic test data from the current workflow build, not mockups. This sample is drawn from a broader 11-scenario test pass covering valid invoices and a range of missing-field, invalid-field, and duplicate cases; not every scenario is pictured individually.

Current Scope & Boundaries

This pipeline handles invoice intake, archival, extraction, validation, duplicate detection, logging, and notification — and is designed to do exactly that, no further.

It is not an ERP, and doesn't replace one. It does not execute or schedule payments. It does not perform accounting approval — nothing here decides an invoice is cleared for payment, only whether it's complete, valid, and new. It is not autonomous accounting: every decision is a visible, rule-based check a finance team can inspect and adjust. And it is not a fraud detection system, nor does it promise error-free processing — it's a first-line layer that gives a reviewer a complete starting point instead of a blank inbox.

Operational Value

The value here isn't any single feature — it's what stays constant, submission after submission.

Consistent validation

every invoice checked against the same rules, regardless of volume

Traceable outcomes

every processed document routed to either the AP Invoice Register or the AP Exception Log

Duplicate control

a resubmitted invoice is checked against the register before it can create a second entry

Original-document access

the archived source file is attached to every outcome, not just a description of it

Structured exceptions

a failed document arrives with its raw OCR text and a named failure reason already attached

Human review where automation should stop

nothing here approves a payment; the workflow's job ends at a complete starting point for a person to act on

Technical Highlights

- Archival and OCR run as parallel branches from a single downloaded attachment — not a sequential chain.
- AI is scoped to structured extraction only; every accept, reject, and duplicate decision is made by deterministic logic.
- Validation returns a named, field-specific failure reason instead of a generic error.
- Duplicate detection runs against the AP Invoice Register, after validation and before a record is written.
- A dual-tab structure keeps confirmed records and exceptions separate, so the register stays clean and the audit trail stays complete.

Potential Extensions

The current build is deliberately scoped. The same architecture is built to extend into these areas — none of which are part of the current implementation:

Direct integration with an accounting or ERP system, replacing the spreadsheet as the system of record

Purchase order matching, to confirm an invoice against an approved PO before logging

Vendor master validation, to flag invoices from vendors outside an approved list

Line-item level extraction, capturing individual charges rather than totals alone

Multi-currency and multi-language OCR support for non-English or non-USD invoices

Each would extend the existing validation and routing logic rather than replace it.

Let's Build Yours

If your AP process still depends on someone manually reading PDFs and retyping totals into a spreadsheet, this is the kind of system that removes that step without removing oversight. This architecture isn't tied to one vendor list or spreadsheet layout — I adapt it around a business's existing AP tools, approval rules, and record-keeping conventions, rather than asking a business to adapt to a template.

If you want to talk through what that looks like for your process, reach out.

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